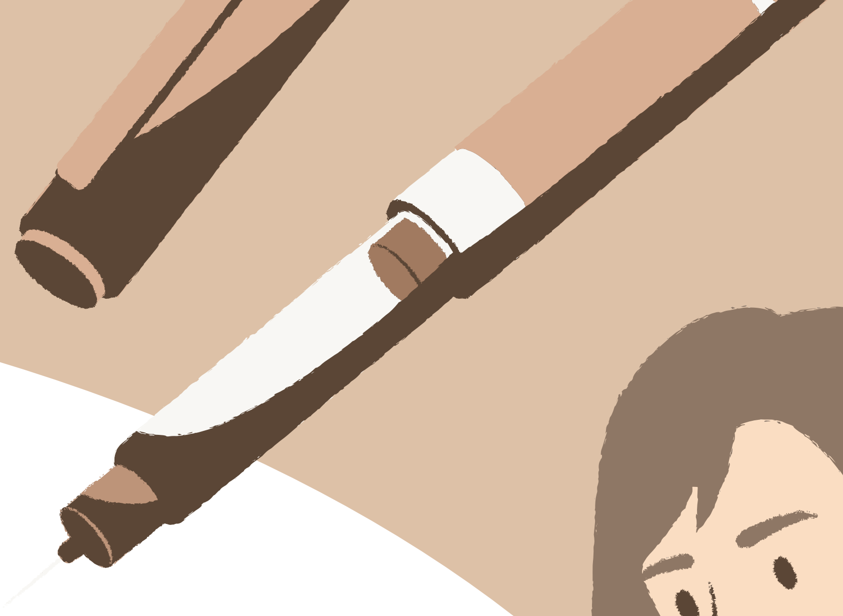
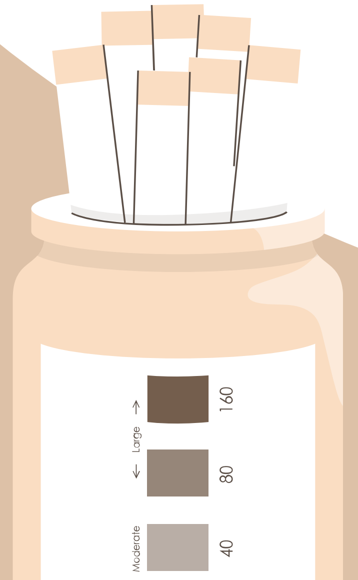
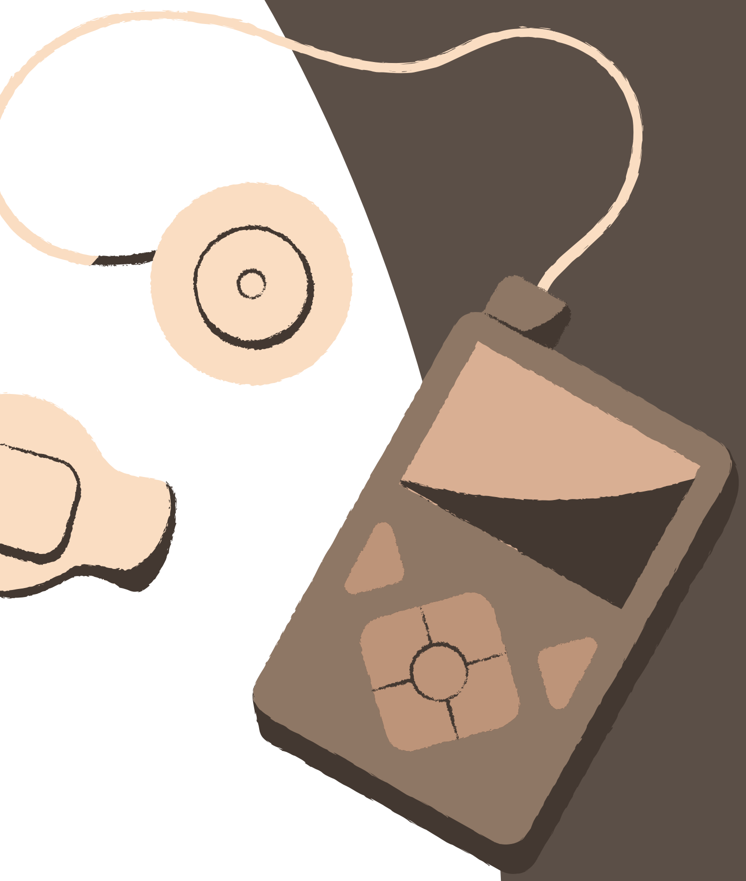


GROUP 8 - EECE 7244

DESIGN AND DEVELOPMENT OF A MEMS-BASED GLUCOSE MONITORING AND AUTOMATED INSULIN DELIVERY SYSTEM

Presented by:
Minh Quach (EECE7244), Shuyue Gao (ME6260), Sri Harish Om Singaravelan Poornima (ME6260), Krishna Shivani (EECE7244), Wentao Yang (ME6260)





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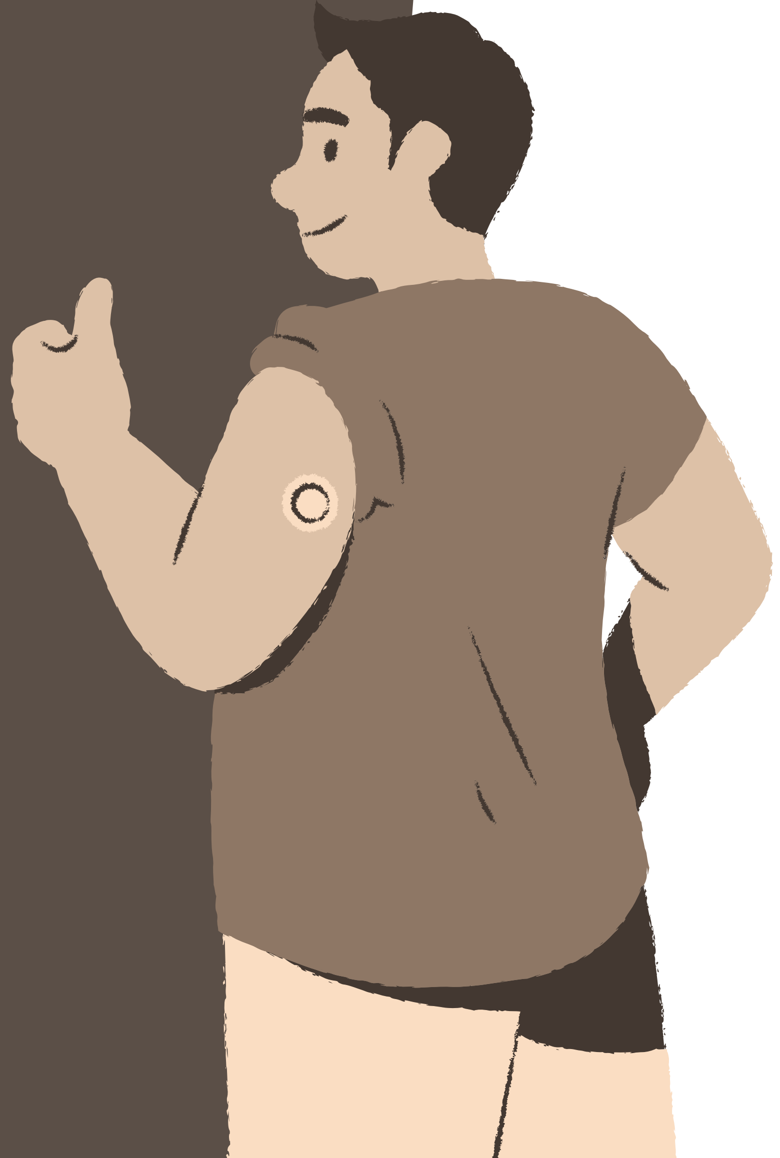


What is diabetes?

Why is glucose monitoring critical?

Role of MEMS technology in addressing gaps?

Introduction

AND OBJECTIVE

Global Diabetes Challenge:
- Over 500 million affected globally.
- Managing glucose levels is critical to avoid severe complications.

Limitations of Current Solutions:
- Invasive, inaccurate, and inconvenient methods.
- Burden of manual insulin dosing.

Why MEMS Technology?
- Real-time glucose monitoring and automated insulin delivery.
- Compact, precise, and user-friendly.

System Overview and Medical Background Motivation

The Problem:

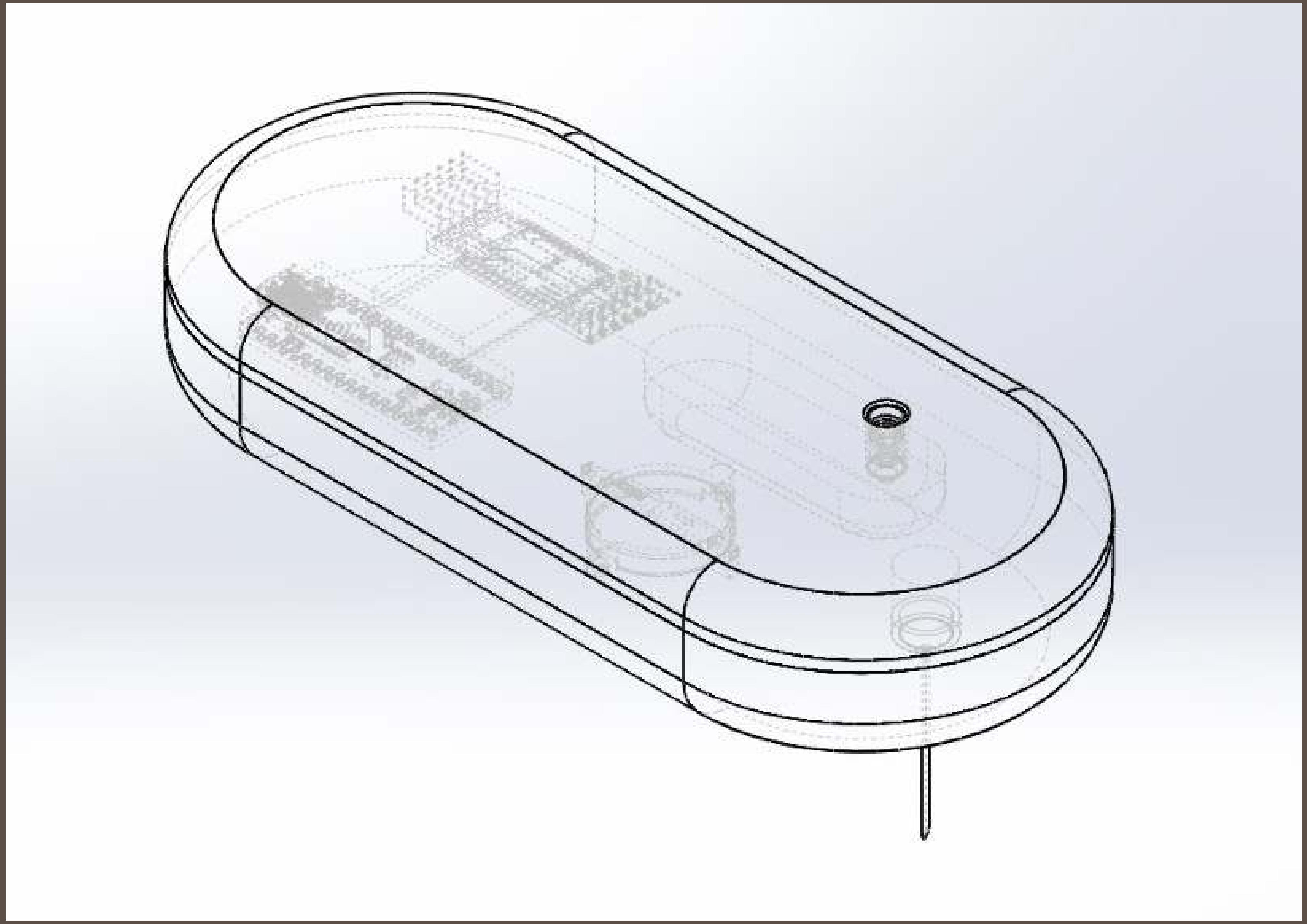
- Traditional methods fail to capture glucose level fluctuations.
- High risk of human error in manual insulin dosing.

Our Solution:

- Real-time, MEMS-based closed-loop system.
- Combines glucose sensors and insulin micropumps

Key Features:

- Continuous monitoring, real-time data analysis, and precise insulin delivery.

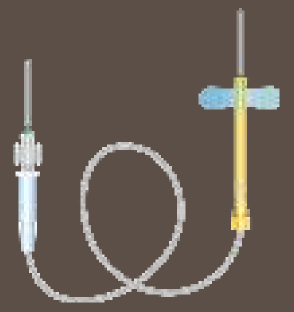


Problem Statement

THE CURRENT METHODS FOR DIABETES MANAGEMENT SUFFER FROM SEVERAL LIMITATIONS:

1. Invasiveness: Traditional glucose monitoring requires frequent finger-prick tests, causing discomfort.
2. Inaccuracy: Delayed or manual insulin dosing can lead to hypo- or hyperglycemia.
3. User Compliance: Cumbersome processes discourage consistent monitoring.

Our solution addresses these issues through a MEMS-based closed-loop system that combines real-time glucose sensing, data processing, and automatic insulin delivery.



Tipos

	OPTICAL SENSOR	ELECTROCHEMICAL SENSORS	SYSTEM MODULE
DESCRIPTION	Non-invasive glucose trend monitoring	High Precision Blood Glucose Testing	Integrated sensing and delivery system, closed loop feedback
CHARACTERISTICS	Real-time data, sensitive to ambient light interference	High sensitivity, miniaturised, requires contact with body fluids	Data fusion, high accuracy, low energy consumption

System Overview

THE MEMS-BASED SYSTEM CONSISTS OF THE FOLLOWING KEY COMPONENTS

MEMS Sensors

Continuous Glucose Monitoring (CGM) sensors measure real-time blood glucose levels.

Microcontroller

Processes sensor data and implements decision-making algorithms.

Insulin Delivery Unit

A micro-pump and micro-needle system administer insulin dynamically.

Wireless Communication

Enables data transfer between sensors, controllers, and mobile applications.

System workflow

1.Blood Glucose Monitoring Device

Input: blood glucose data is collected by the sensor.

Processing: data is processed by ESP32 and sent via Bluetooth module.

Output: Data is transmitted to the insulin delivery system for calculation of insulin dose.

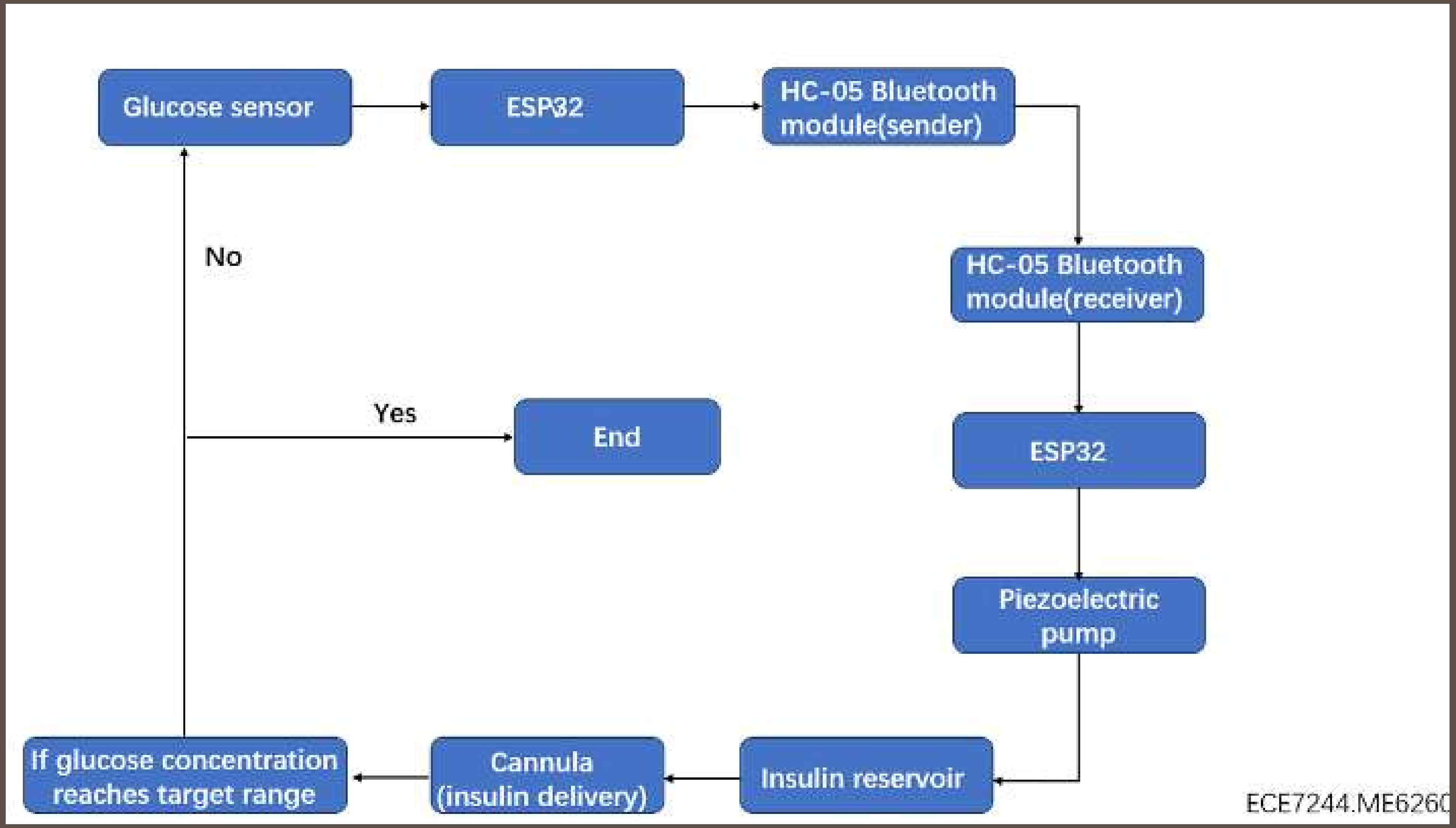
2.Insulin Delivery System

Receiving: Bluetooth module receives blood glucose data.

Calculating: The ESP32 calculates the insulin dose.

Drive: Piezo pump delivers insulin from storage compartment.

Deliver: Insulin is delivered into the body through the injection cannula.



MEMS Technology

- MEMS (Micro-Electro-Mechanical Systems) technology plays a critical role in miniaturizing medical devices.

- Electrochemical Sensors: Measure glucose levels through enzymatic reactions.
- Optical Sensors: Utilize light absorption or fluorescence for non-invasive glucose detection.

Types of MEMS Sensors

- High sensitivity and precision.
- Compact and biocompatible design for wearable devices.

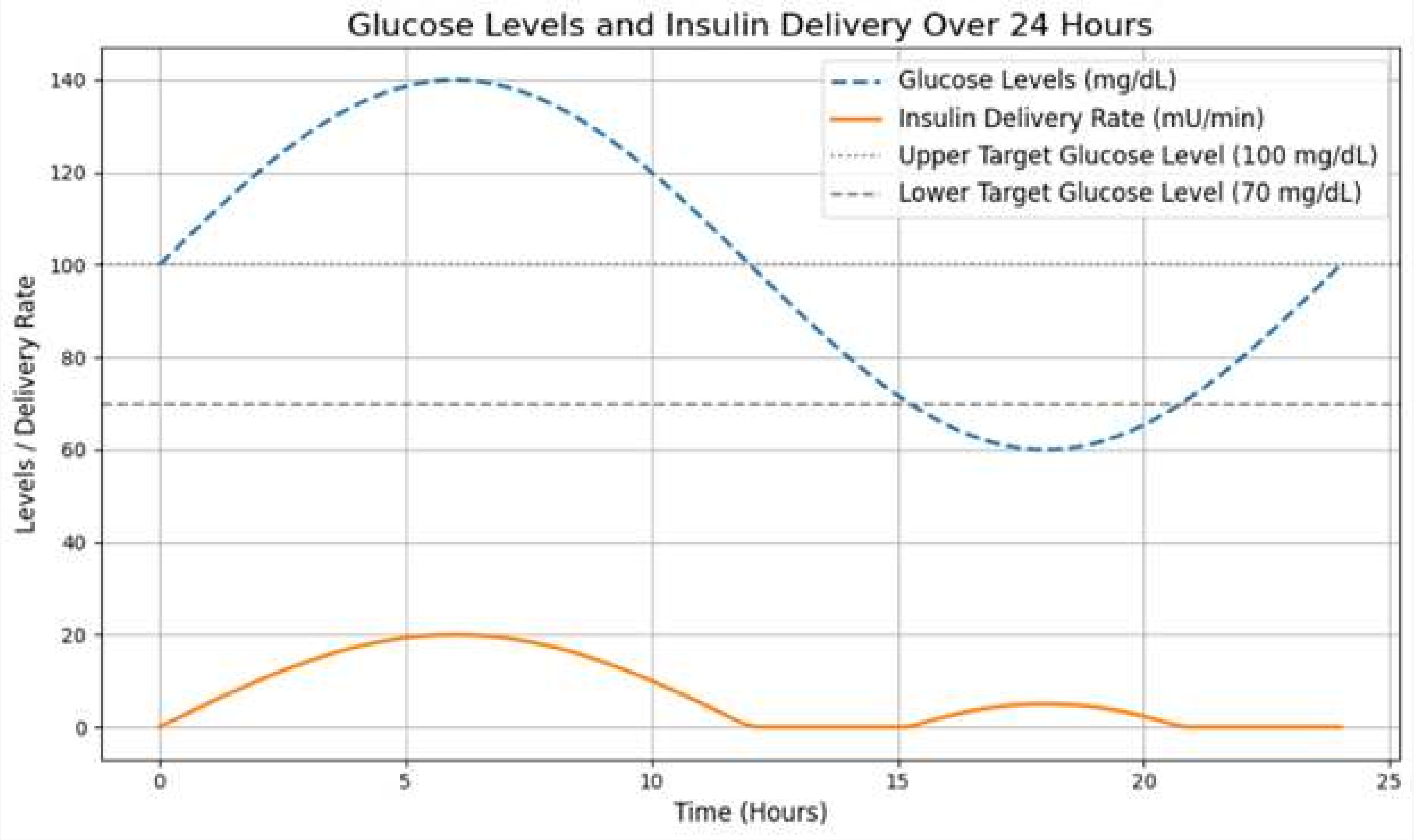
Advantages

Modeling and Simulation for Insulin Delivery System

- **Overview of Mathematical Models:**
  - The system is modeled using the Bergman Minimal Model, which simulates glucose-insulin dynamics:
    - $\frac{dG}{dt} = -P_1 G - x(G + G_b) + D$
    - $\frac{dI}{dt} = -n(I + I_b) + \frac{U}{v_1}$
    - $\frac{dx}{dt} = -P_2 x + P_3 I$
- **PID Control Mechanism:**
  - A Proportional-Integral-Derivative (PID) controller dynamically adjusts insulin delivery based on real-time glucose fluctuations.

$$u = U_{ff} + k_p(G - G_{sp}) + k_d \frac{dG}{dt} + k_e \int_0^{\infty} (G - G_{sp})\, dt$$

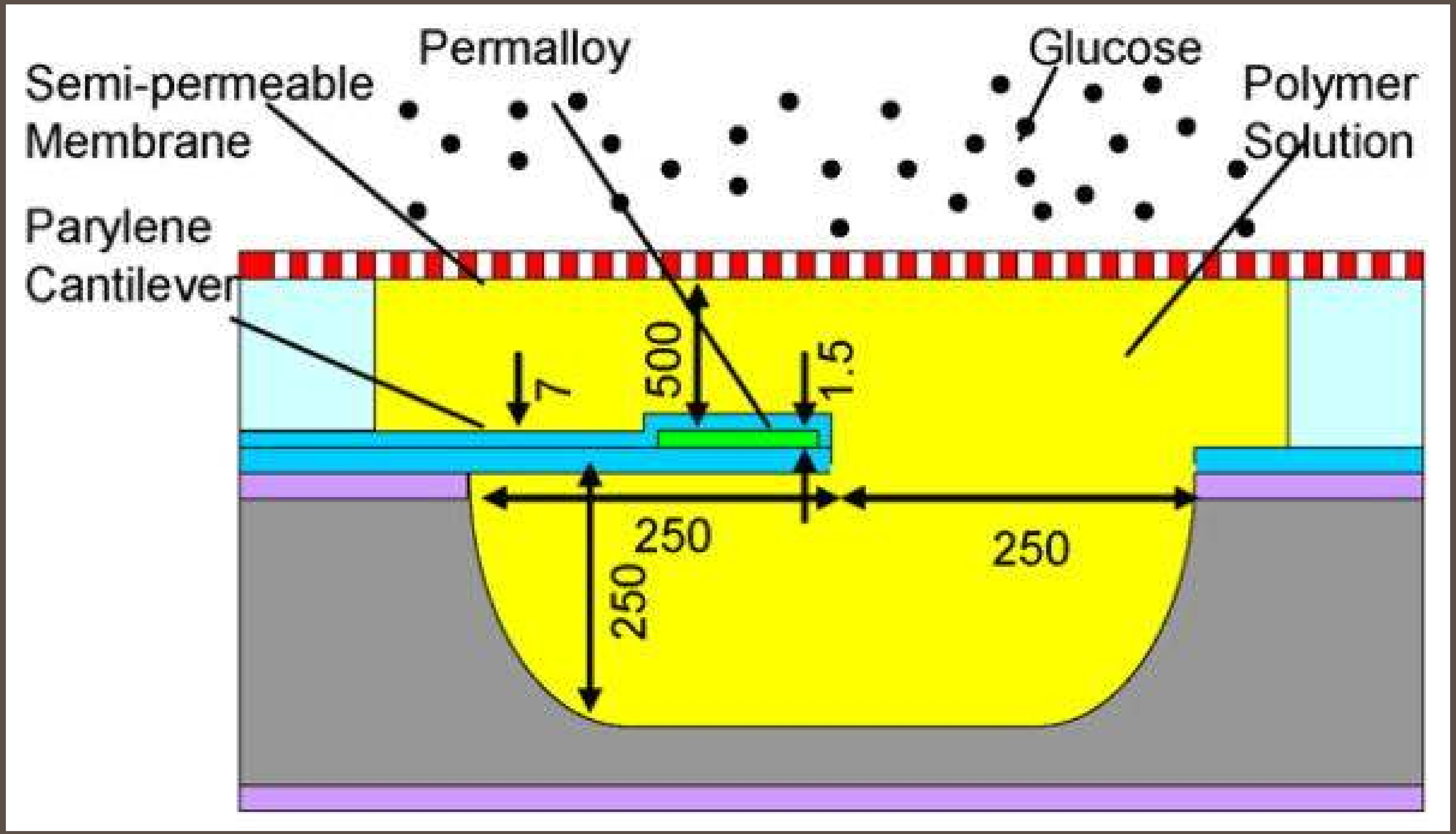
```python
import matplotlib.pyplot as plt
import numpy as np

# Simulated data for glucose levels and insulin delivery rates
time = np.linspace(0, 24, 100)  # Simulating 24 hours
# Updated glucose level simulation with wider variation
glucose_levels = 100 + 40 * np.sin(2 * np.pi * time / 24)  # Sinusoidal variation for glucose levels

# insulin delivery logic to include both thresholds (<70 and >100 mg/dL)
insulin_delivery = np.where(glucose_levels < 70, 0.5 * (70 - glucose_levels),
                            np.where(glucose_levels > 100, 0.5 * (glucose_levels - 100), 0))

# Plotting glucose levels vs insulin delivery rates
plt.figure(figsize=(10, 6))
plt.plot(time, glucose_levels, label='Glucose Levels (mg/dL)', linestyle='--', linewidth=2)
plt.plot(time, insulin_delivery, label='Insulin Delivery Rate (mU/min)', linewidth=2)
plt.axhline(y=100, color='gray', linestyle=':', label='Upper Target Glucose Level (100 mg/dL)')
plt.axhline(y=70, color='gray', linestyle='--', label='Lower Target Glucose Level (70 mg/dL)')

# Adding labels and legend
plt.title('Glucose Levels and Insulin Delivery Over 24 Hours', fontsize=16)
plt.xlabel('Time (Hours)', fontsize=12)
plt.ylabel('Levels / Delivery Rate', fontsize=12)
plt.legend(fontsize=12)
plt.grid(True)
plt.tight_layout()
plt.show()
```



MEMS Sensor

MEMS Glucose Sensor

MEMS affinity CGM sensor design. A cantilever vibrates in a glucose-sensitive polymer solution inside a microchamber. Glucose permeates through a semi-permeable membrane, changing the solution viscosity and vibration damping. (Dimensions for a prototype device are given in micrometers).



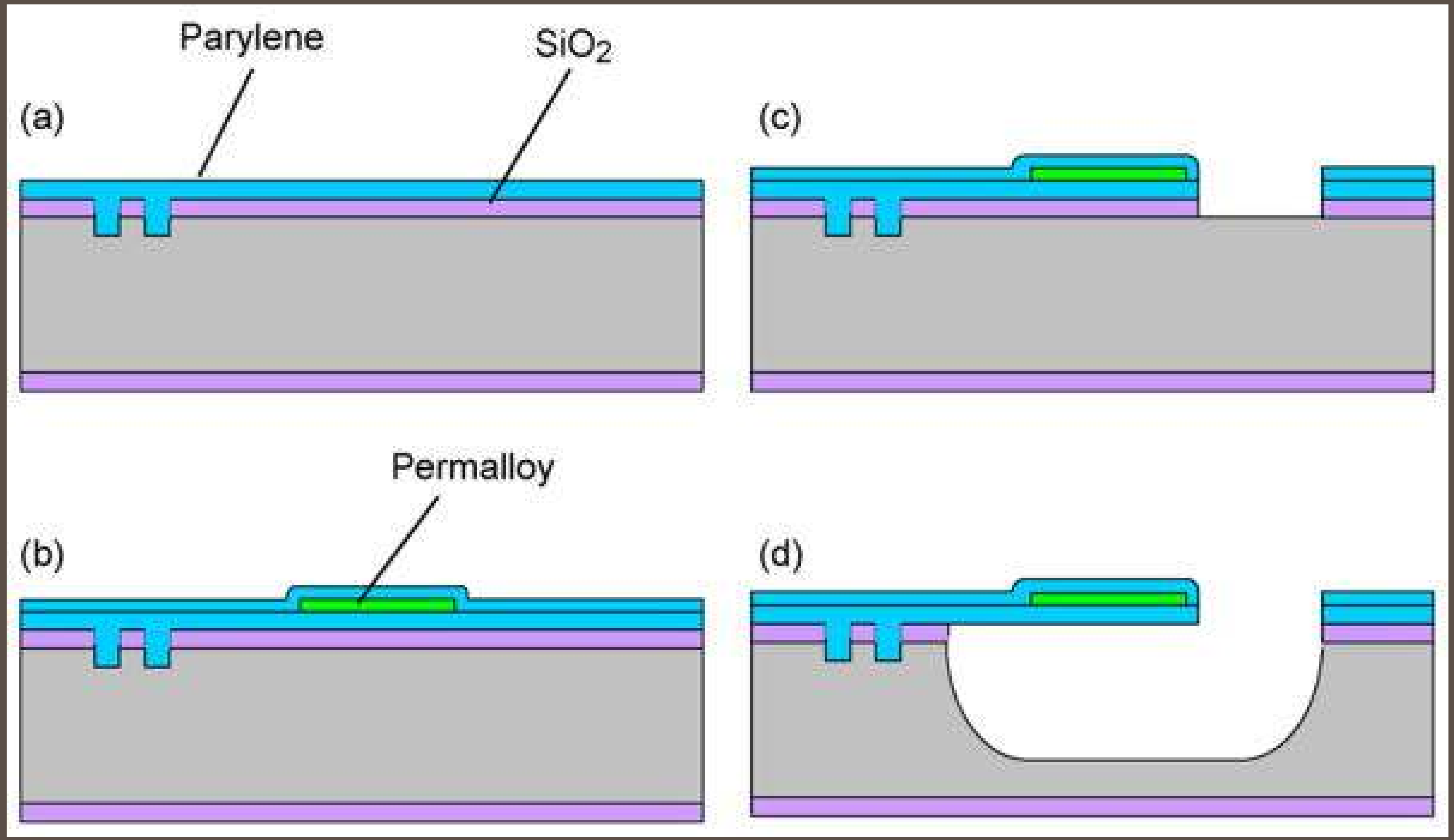
Fabrication Process

(a) a Parylene layer (5 μm) was deposited by chemical vapor deposition onto a SiO2-coated silicon wafer into which small cavities were etched.

(b) A 100 nm copper seed layer was then deposited on the Parylene, followed by the deposition and patterning a 1.5 μm photoresist layer defining the permalloy deposition area (150 μm long and 200 μm wide). A permalloy thin film was then deposited by electroplating, followed by the removal of the photoresist and unused copper, and the deposition of a second Parylene layer (2 μm).

(c) The two Parylene layers, along with the underlying SiO2 layer, were then patterned to define a cantilever (250 μm in both length and width).

(d) The cantilever was finally released by gas-phase XeF2 etching of the silicon underneath (forming a cavity approximately 500 μm × 500 μm × 250 μm in dimensions) and removal of the SiO2 directly beneath the cantilever.



Microcontroller

Feature	ESP32	Arduino
CPU	Dual-core (up to 240 MHz)	Single-core (16 MHz on Uno)
Memory	Up to 520 KB SRAM, 4 MB Flash	2 KB SRAM, 32 KB Flash (Uno)
Connectivity	Wi-Fi, Bluetooth, BLE	None (native), shields required for expansion
I/O Pins	Up to 34 GPIO pins	14 digital I/O pins and 6 analog (Uno)
Analog Inputs	Up to 18 channels	6 channels (Uno)
Voltage	3.3V	5V (Uno)
Power Consumption	Low power modes available (deep sleep, etc.)	Low power modes available but less efficient
Integrated Features	Hall sensor, temperature sensor, touch sensors	None (native)
Price	Comparable to or lower than Arduino (pack of three for the same)	Generally low cost
Programming Environment	ESP-IDF, Arduino IDE (with add-on)	Arduino IDE

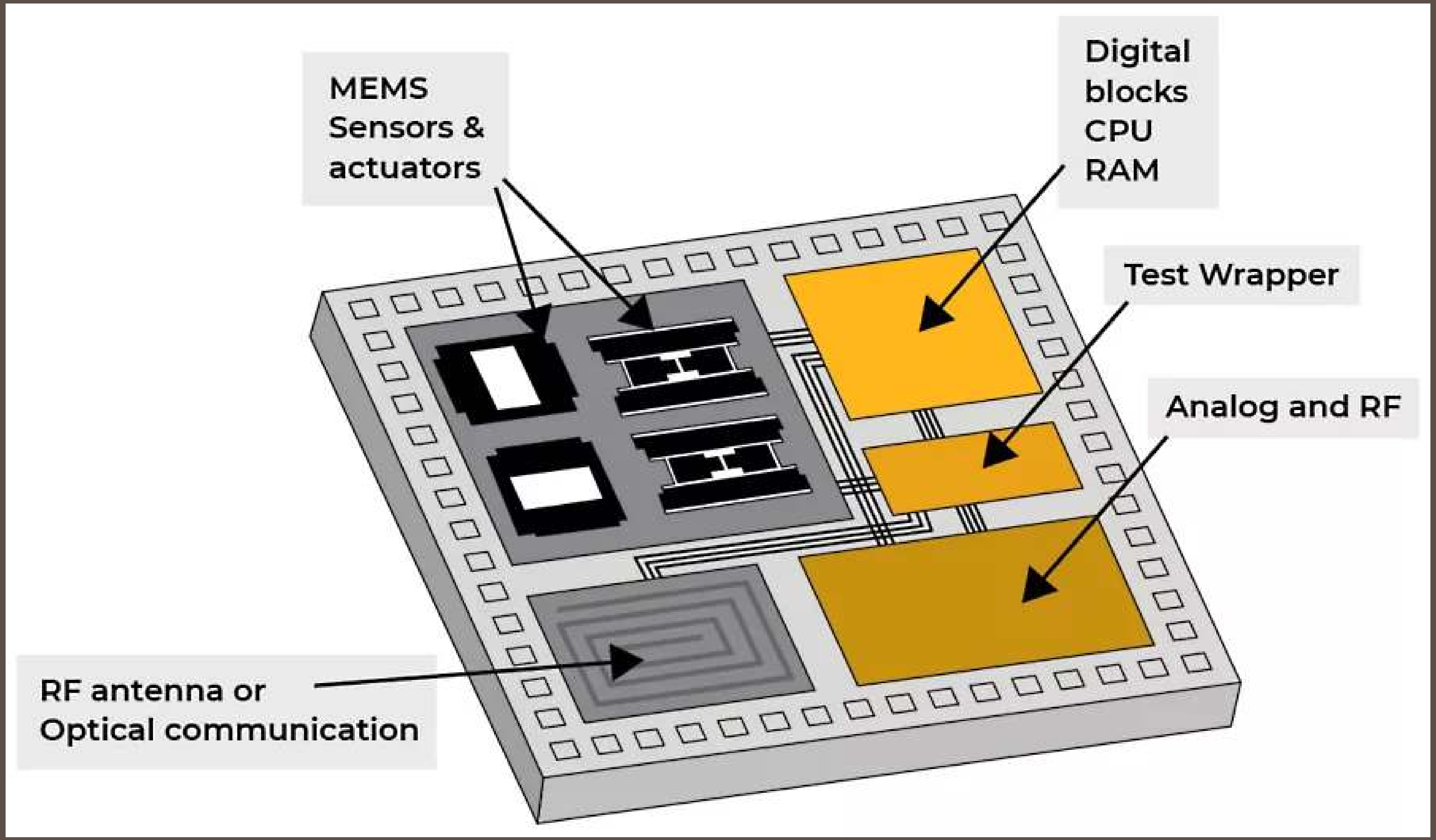
ESP32

The ESP32 is a highly versatile, low-cost SoC(System on Chip) chip that has a built-in microprocessor, a full TCP/IP protocol stack, and can connect straight to your Wi-Fi network.

Microcontroller

COMPONENTS CONTAINED WITHIN AN SOC

- DATA PROCESSING UNITS
- EMBEDDED MEMORY
- GRAPHICS PROCESSING UNITS(GPU)
- USB INTERFACES
- VIDEO AND AUDIO PROCESSING



Insulin delivery unit

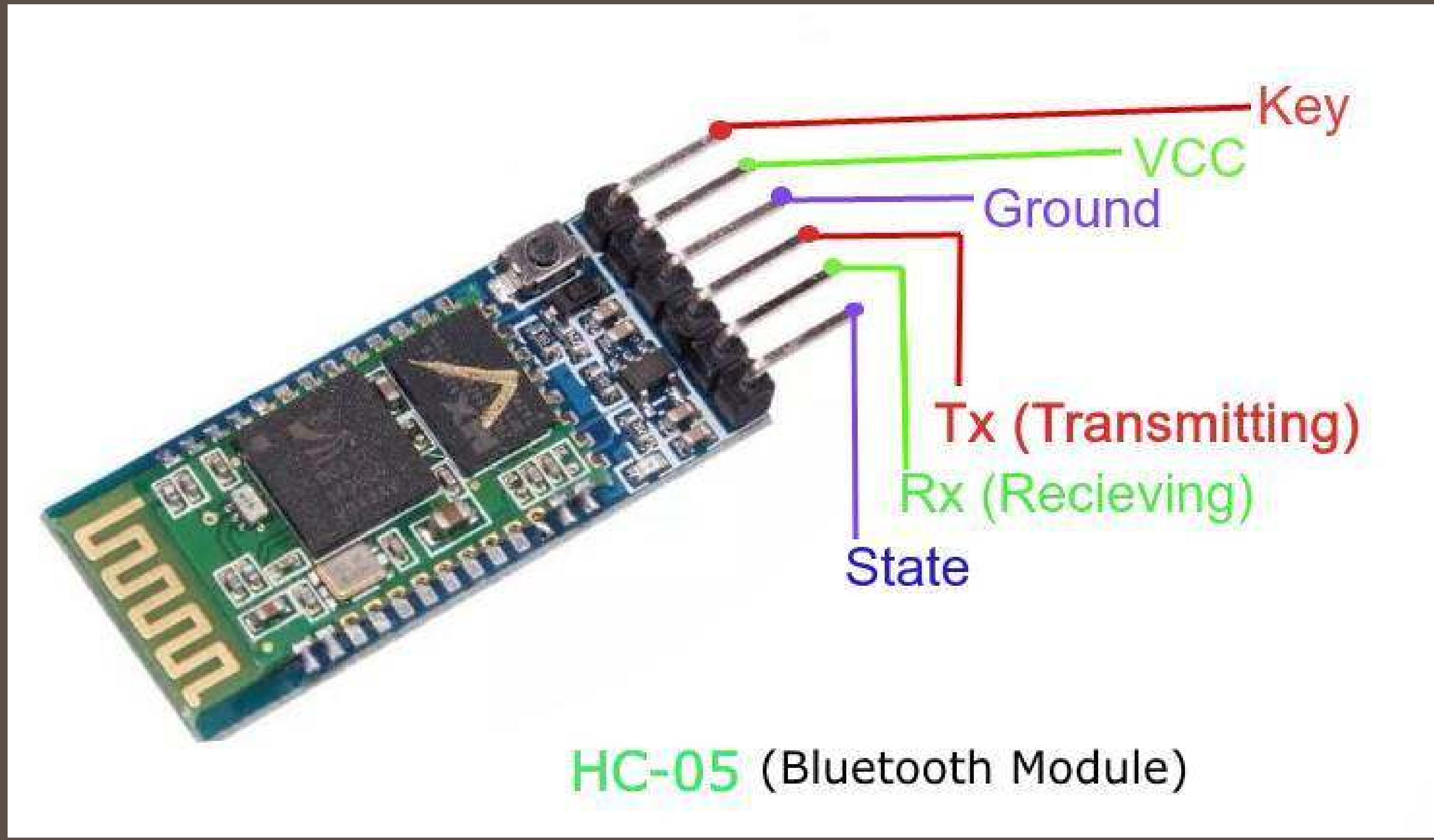
- IMPLEMENTING AN INSULIN PUMP SYSTEM WITH THE ESP32 INVOLVES A COMPREHENSIVE STRATEGY PRIORITIZING ACCURACY, RELIABILITY, AND SAFETY.
- THE ESP32 INTEGRATES SEAMLESSLY WITH GLUCOSE SENSORS TO MONITOR REAL-TIME BLOOD GLUCOSE LEVELS AND CALCULATE PRECISE INSULIN DOSES, ENABLING DYNAMIC ADJUSTMENTS TO MEET USERS' CHANGING PHYSIOLOGICAL NEEDS.
- ADDITIONALLY, WITH BLUETOOTH, WHICH ENSURES ROBUST COMMUNICATION WITHIN THE SYSTEM AND ENHANCES THE USER EXPERIENCE,
- ESP32 IS EQUIPPED WITH MULTIPLE PROTECTION LAYERS TO PREVENT OVERDOSES AND ENSURE RELIABLE PERFORMANCE UNDER VARIOUS CONDITIONS.
- OPTIMIZED POWER MANAGEMENT EXTENDS BATTERY LIFE AND ENSURES UNINTERRUPTED OPERATION, CATERING TO THE PORTABILITY NEEDS OF DAILY INSULIN PUMP USERS.

Wireless Communication

Bluetooth is a short-range wireless technology standard that is used for exchanging data between fixed and mobile devices over short distances and building personal area networks (PANs). In the most widely used mode, transmission power is limited to 2.5 milliwatts, giving it a very short range of up to 10 metres (33 ft). It employs UHF radio waves in the ISM bands, from 2.402 GHz to 2.48 GHz.

In this program, the HC-05 bluetooth module is used to deliver and receive signals.

Bluetooth®

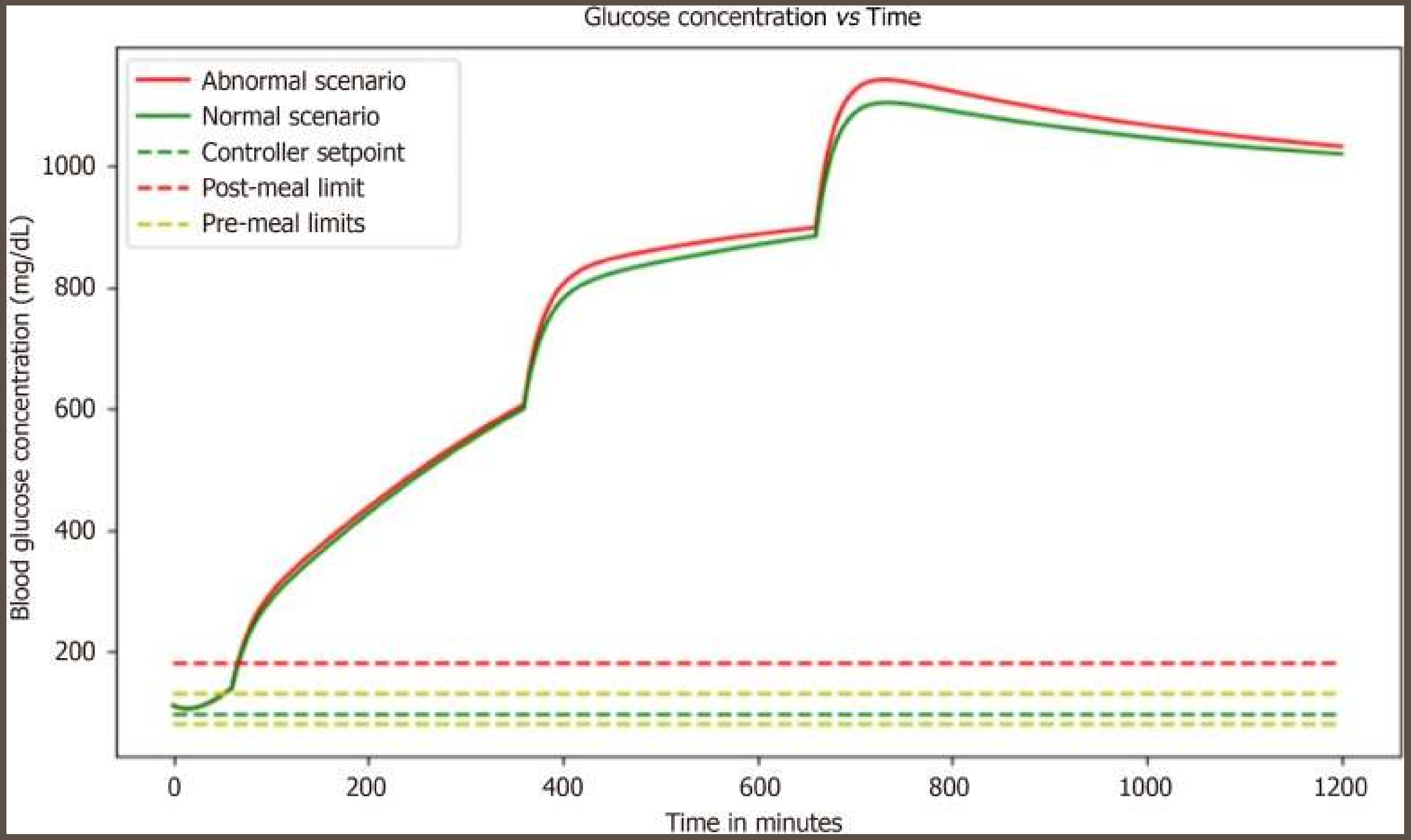


HC-05

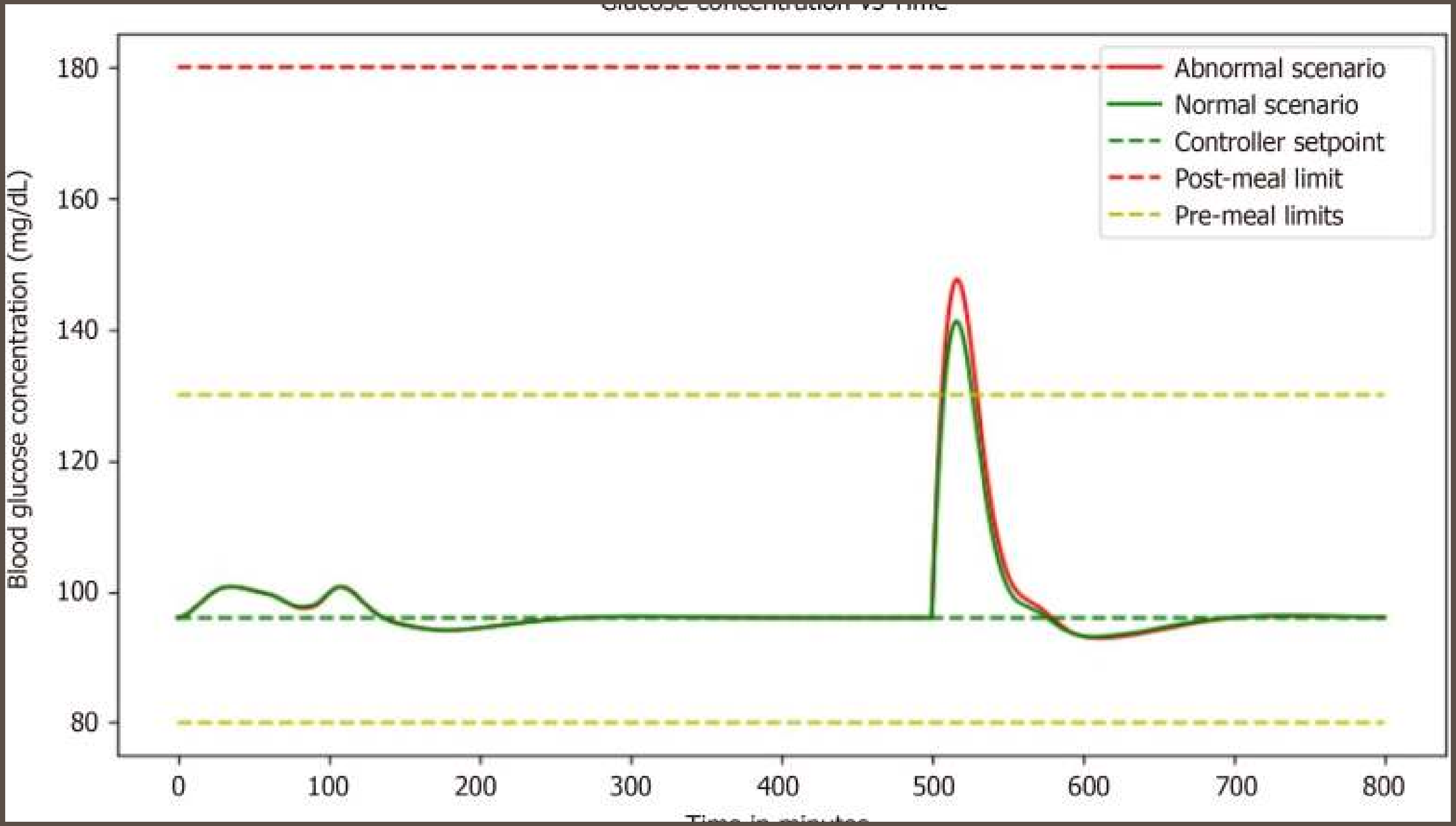
System Integration and Control

- THE SYSTEM FEATURES AN AI DETECTION BLOCK THAT ANALYZES GLUCOSE DATA USING ADVANCED ALGORITHMS AND MACHINE LEARNING, ENABLING INFORMED INSULIN DOSING RECOMMENDATIONS. THIS CLOSED-LOOP DESIGN COMBINES REAL-TIME MONITORING WITH AI-DRIVEN DECISION-MAKING, REDUCING THE NEED FOR MANUAL INTERVENTION.
-
- A GLUCOSE SENSOR COLLECTS DATA NON-INVASIVELY OR MINIMALLY INVASIVELY, ELIMINATING TRADITIONAL FINGER TESTS. THE DATA IS SEAMLESSLY PROCESSED BY A MACHINE LEARNING MODEL TO INTERPRET COMPLEX GLUCOSE PATTERNS.
-
- THE MODEL IDENTIFIES PATTERNS AND ANOMALIES IN GLUCOSE LEVELS INFLUENCED BY FACTORS SUCH AS DIET, ACTIVITY, AND STRESS, OFFERING A PERSONALIZED RESPONSE FOR EACH USER.
-
- THE SYSTEM EMPLOYS THE BERGMAN MINIMAL MODEL AND A PID CONTROLLER TO MANAGE GLYCEMIC CONTROL, MAINTAINING BLOOD GLUCOSE AT A DESIRED SETPOINT.
-
- INSULIN DOSAGE ADJUSTMENTS ARE CALCULATED BY THE CONTROL SYSTEM AND ADMINISTERED VIA A MICROPUMP AND MICRO-NEEDLE, ENSURING PRECISE AND ADAPTIVE INSULIN DELIVERY.

Conclusion

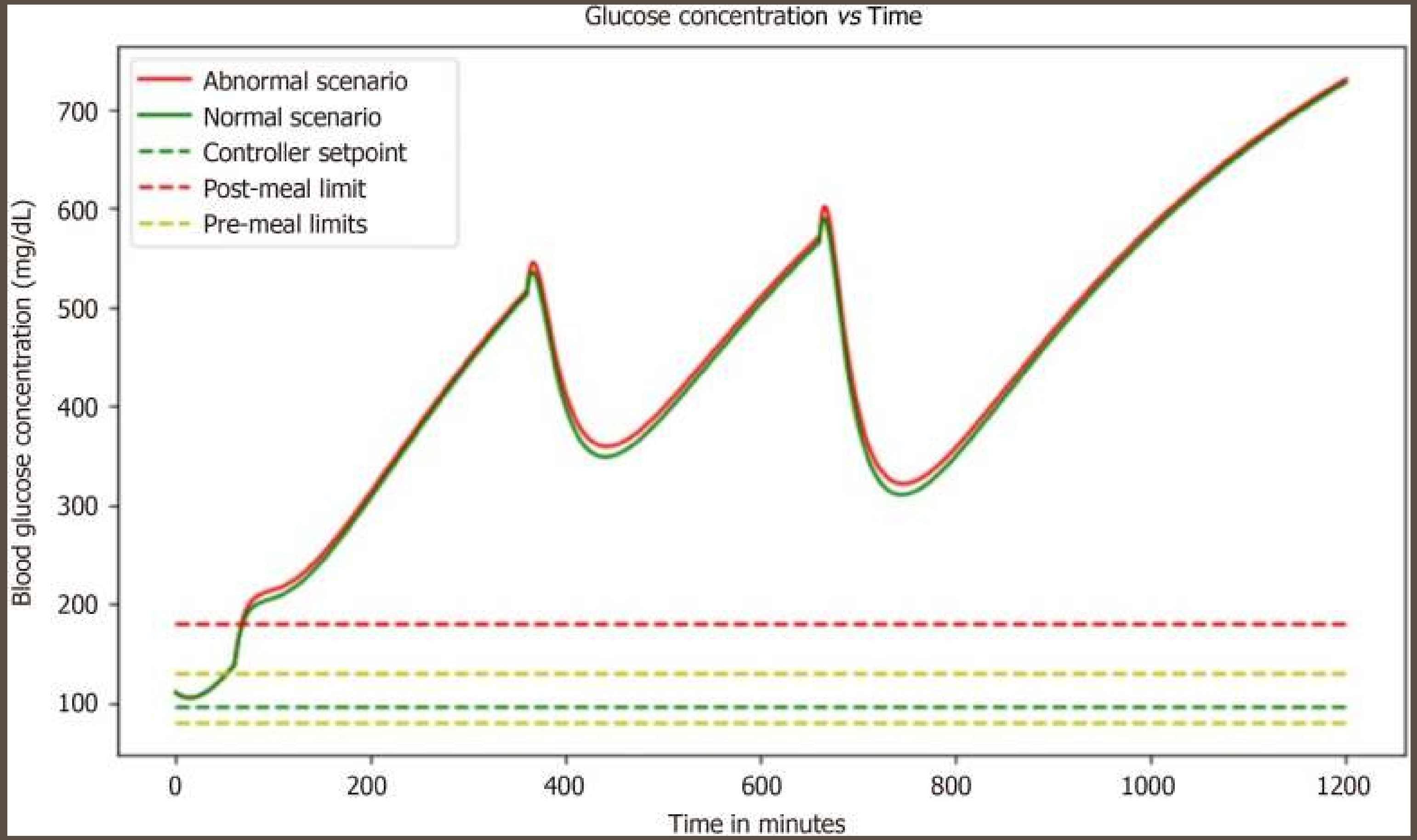


Graph depicts the dynamic relationship between glucose concentration and time, revealing fluctuations that highlight the critical need for precise glucose management in diabetes care.

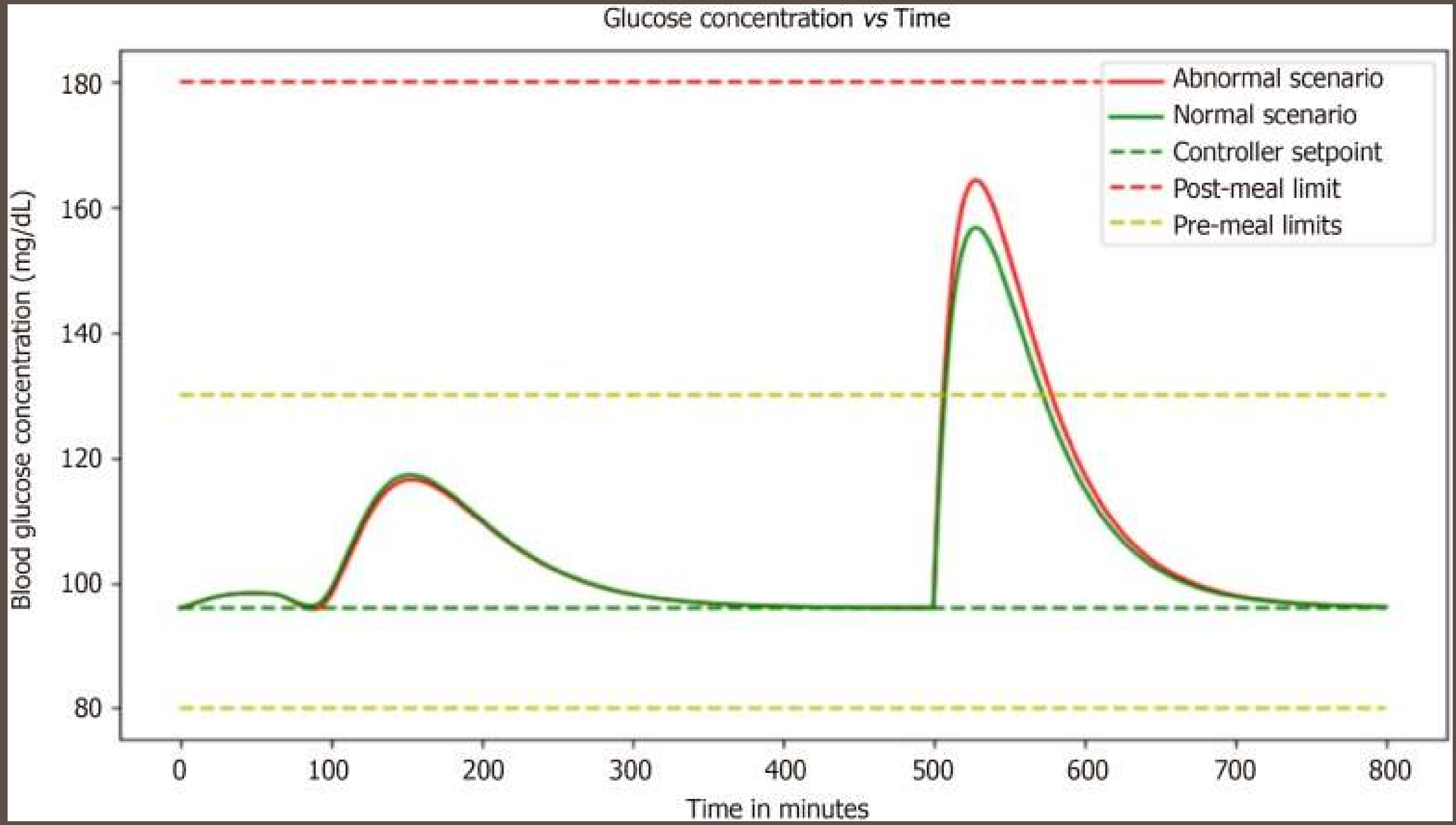


Graph showcases the correlation between glucose concentration and basal insulin dosage, providing insights into the effectiveness of the basal insulin regimen in maintaining stable glucose levels.

Conclusion

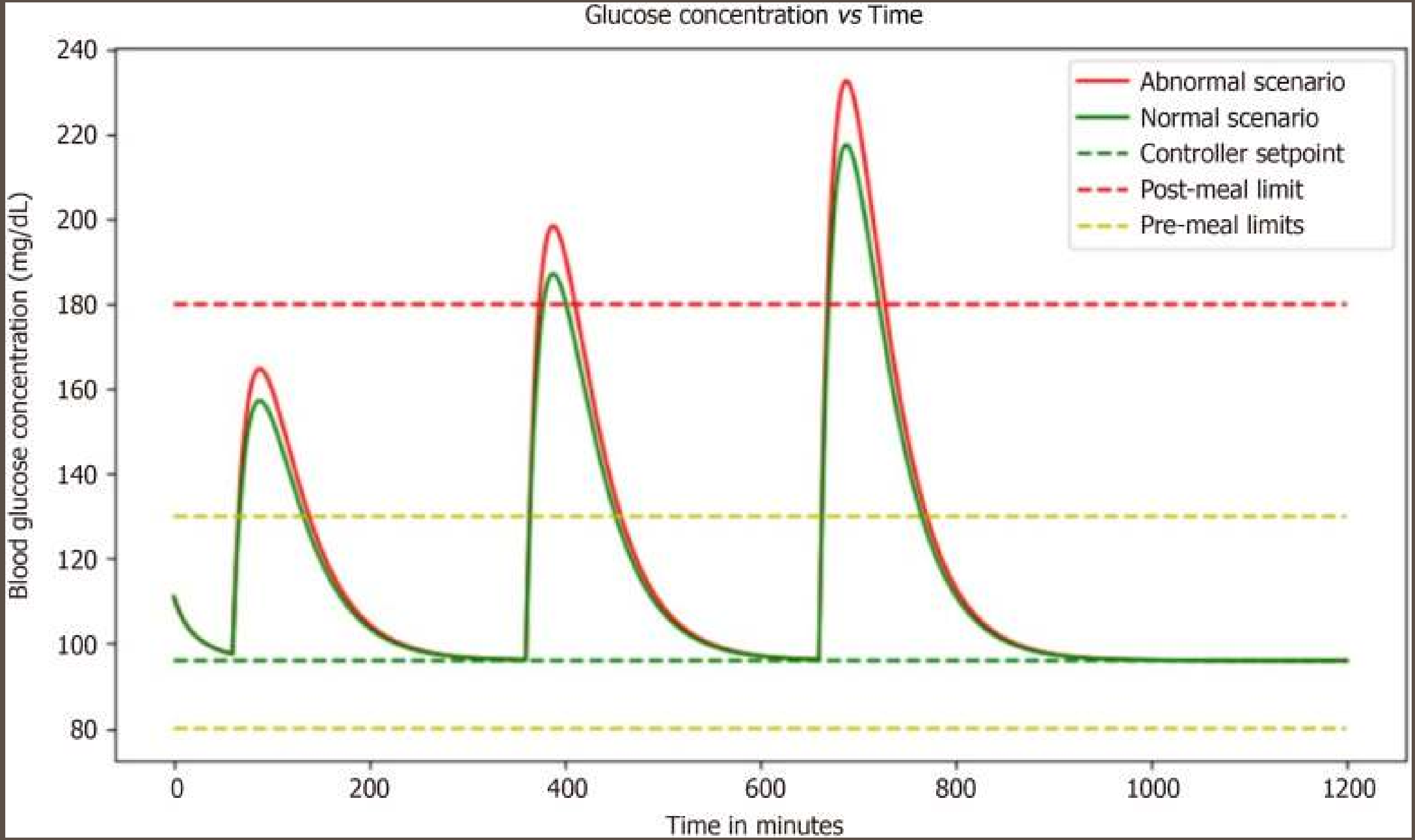


Graph highlights the impact of high glucose concentrations, revealing the critical importance of effective interventions to prevent hyperglycemia-related complications in diabetes management, but after a certain period of time glucose levels increases abnormally.

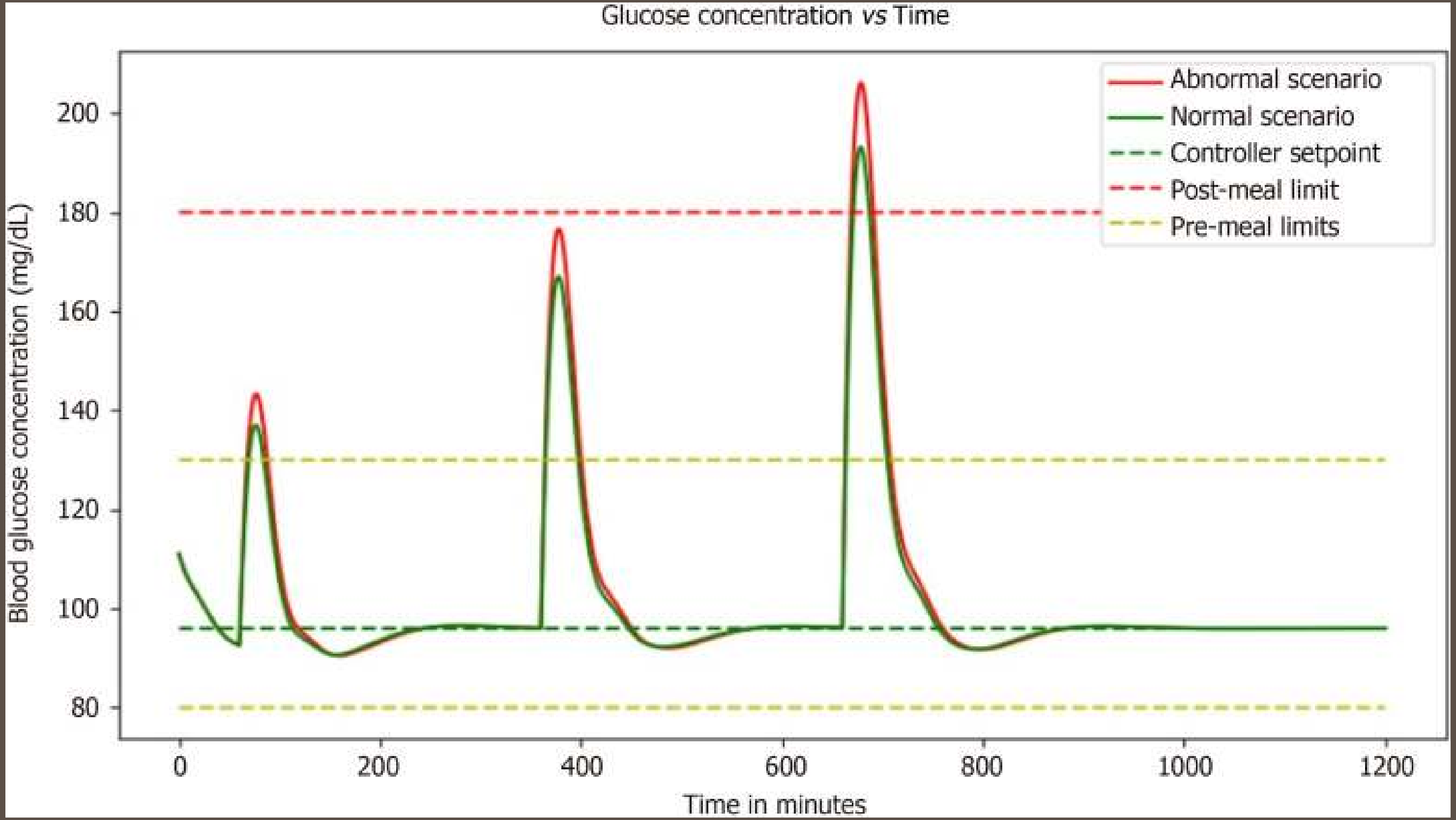


Graph illustrates the dynamic interaction between meal intake and basal insulin dose, showcasing the intricate balance required for optimal glucose control in diabetes management.

Results



Graph demonstrates the performance of the proportional-integral-derivative controller, showcasing its ability to efficiently regulate a system's output by continuously adjusting the control parameters.



Graph illustrating the combined effect of proportional-integral-derivative controller with basal dose which is measured to evaluate the ability to effectively regulate glucose levels.

Future Enhancement

Enhanced Algorithms and Application

Incorporating AI models for predictive analysis and trend detection and xtending the system to manage other chronic conditions

Improved Materials

Utilizing advanced biocompatible materials for longer device lifespans. These advancements will further optimize system performance and broaden its applications in healthcare.

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THANK YOU!

Os esperamos
en nuestra clínica